

Task 5 - Exploratory Data Analysis (EDA) Report

Objective

Perform Exploratory Data Analysis (EDA) on the Titanic dataset to identify patterns, trends, missing values, outliers, and relationships using statistical summaries and visualizations.

Dataset

Titanic Dataset (891 records, 12 features).

Analysis Performed

- Loaded the dataset using Pandas.
- Explored the data using `head()`, `info()`, `describe()`, and `value_counts()`.
- Created Histogram, Boxplot, Scatter Plot, Correlation Heatmap, and Pairplot.
- Recorded observations for each visualization.

Key Findings

- Missing values were found in Age, Cabin, and Embarked.
- Most passengers were young adults.
- Male passengers outnumbered female passengers.
- Fare contains several high-value outliers.
- There is no strong relationship between Age and Fare.
- Most numerical features showed weak correlations.
- Higher-class passengers and those who paid higher fares generally had better survival rates.

Conclusion

The analysis provided meaningful insights into passenger characteristics and survival patterns and demonstrated the importance of EDA before building predictive models.